

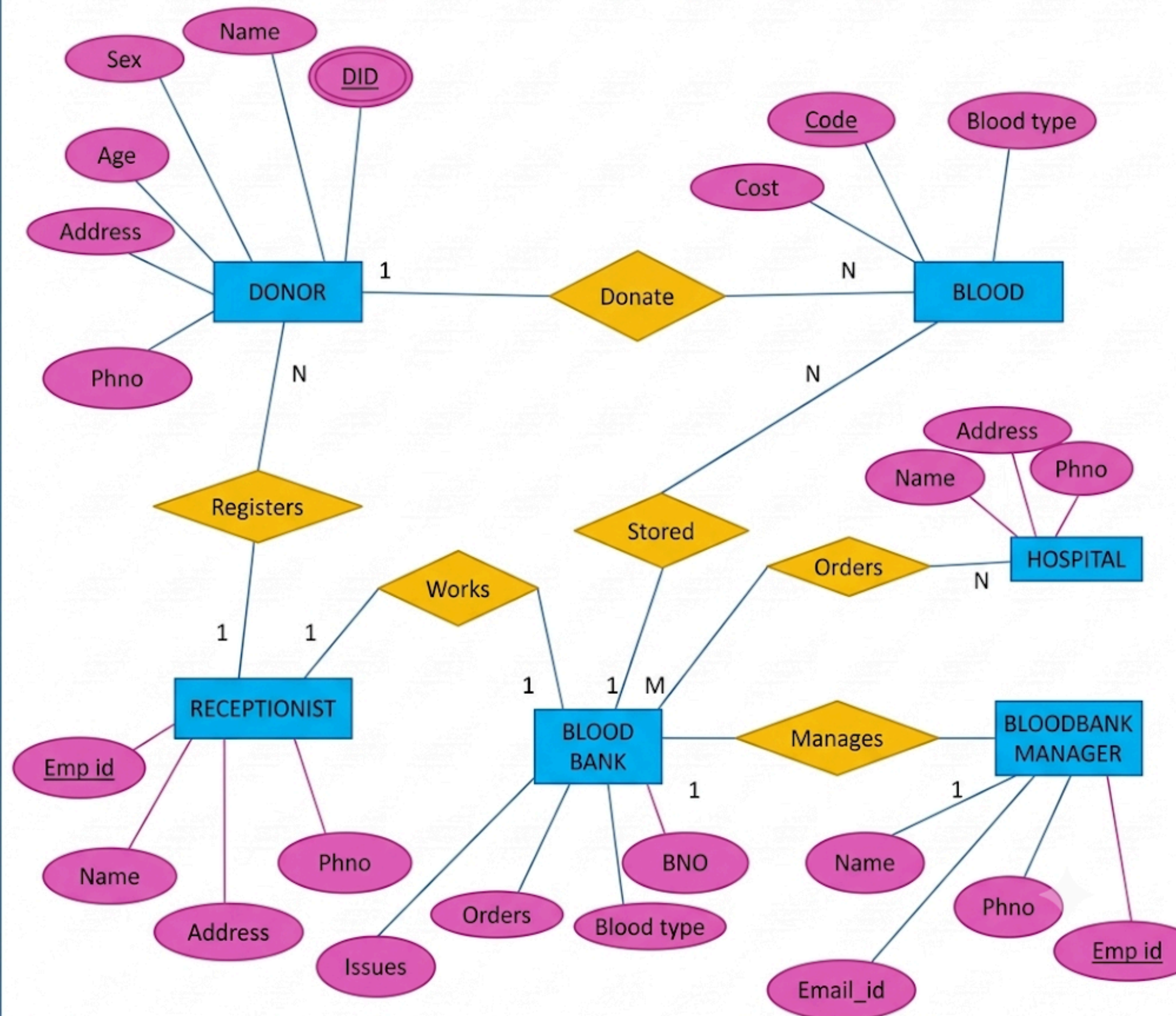
DATABASE MANAGEMENT SYSTEM • PHASE 2

Blood Bank Donation & Issue Management

Design and Implementation using MySQL

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THE COMPLETE ER - DIAGRAM



Relational schema overview

Seven normalized tables organize people, inventory, events, and institutions.

Donor

Donor_ID PK • Name • Blood_Type • Contact

Blood_Bank

BB_ID PK • Name • Location

Blood_Unit

Unit_ID PK • BB_ID FK • Blood_Type • Expiry_Date • Status

Donation

Donation_ID PK • Donor_ID FK • Unit_ID FK • Donation_Date

Patient

Patient_ID PK • Name • Blood_Type • Contact

Hospital

Hospital_ID PK • Name • Location

Blood_Issue

Issue_ID PK • Unit_ID FK • Patient_ID FK • Hospital_ID FK • Issue_Date

Table structures — core inventory

Part 1: donor identity, blood-bank location, and blood-unit status.

DONOR FIELD		TYPE / KEY		BANK FIELD		TYPE / KEY		UNIT FIELD		TYPE / KEY	
Donor_ID		INT • PK		BB_ID		INT • PK		Unit_ID		INT • PK	
Name		VARCHAR		Name		VARCHAR		BB_ID		INT • FK	
Blood_Type		CHAR		Location		VARCHAR		Blood_Type		CHAR	
Contact		VARCHAR						Expiry_Date		DATE	
								Status		VARCHAR	

Table structures — transactions & recipients

Part 2: donation events, recipients, hospitals, and issue transactions.

DONATION FIELD		TYPE / KEY
Donation_ID		INT • PK
Donor_ID		INT • FK
Unit_ID		INT • FK
Donation_Date		DATE

PATIENT FIELD		TYPE / KEY
Patient_ID		INT • PK
Name		VARCHAR
Blood_Type		CHAR
Contact		VARCHAR

HOSPITAL FIELD		TYPE / KEY
Hospital_ID		INT • PK
Name		VARCHAR
Location		VARCHAR

ISSUE FIELD		TYPE / KEY
Issue_ID		INT • PK
Unit_ID / Patient_ID / Hospital_ID		INT • FK
Issue_Date		DATE

Sample records — donors & blood banks

Realistic test data verifies identifiers, contact values, and location fields.

```
mysql> SELECT * FROM Donor;
+-----+-----+-----+-----+-----+-----+-----+
| Donor_ID | Name       | Age | Gender | Blood_Group | Phone       | Adress       |
+-----+-----+-----+-----+-----+-----+-----+
| 101      | Rahul Sharma | 24  | Male   | O+          | 9876543210 | Hyderabad   |
| 102      | Ananya Roy   | 22  | Female | A+          | 9876543211 | Hyderabad   |
| 103      | Arjun Das    | 29  | Male   | B+          | 9876543212 | Secunderabad |
| 104      | Priya Singh  | 26  | Female | AB+         | 9876543213 | Hyderabad   |
| 105      | Rohit Kumar  | 31  | Male   | O-          | 9876543214 | Warangal     |
+-----+-----+-----+-----+-----+-----+-----+
5 rows in set (0.04 sec)
```

Sample records — inventory & donations

Each donation points to a specific donor and blood unit; inventory records expose availability.

```
mysql> SELECT * FROM Donation;
+-----+-----+-----+-----+-----+-----+
| Donation_ID | Donor_ID | Blood_ID | Donation_Date | Quantity | Status |
+-----+-----+-----+-----+-----+-----+
| 601 | 101 | 501 | 2026-09-01 | 450 | Approved |
| 602 | 102 | 502 | 2026-09-03 | 450 | Approved |
| 603 | 103 | 503 | 2026-09-05 | 450 | Approved |
| 604 | 104 | 504 | 2026-09-06 | 450 | Approved |
| 605 | 105 | 505 | 2026-09-08 | 450 | Approved |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.04 sec)
```

```
mysql> SELECT * FROM Blood_Issue;
+-----+-----+-----+-----+-----+-----+
| Issue_ID | Patient_ID | Blood_ID | Hospital_ID | Issue_Date | Quantity |
+-----+-----+-----+-----+-----+-----+
| 701 | 201 | 501 | 301 | 2026-09-10 | 450 |
| 702 | 202 | 502 | 302 | 2026-09-11 | 450 |
| 703 | 203 | 503 | 303 | 2026-09-12 | 450 |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

SQL demonstrations — UPDATE & JOIN

Change inventory state, then combine donor identity with donation history.

```
UPDATE Blood_Unit SET Status = 'Used'
WHERE Unit_ID = 1001;
```

Result: Unit 1001 → Used

```
SELECT Donor.Name, Donation.Donation_Date
FROM Donor JOIN Donation
ON Donor.Donor_ID = Donation.Donor_ID;
```

```
mysql> SELECT * FROM Blood_Unit;
```

Blood_ID	Blood_Group	Collection_Date	Expiry_Date	Quantity	Status	Bank_ID
501	O+	2026-09-01	2026-10-13	450	Available	1
502	A+	2026-09-03	2026-10-15	450	Available	1
503	B+	2026-09-05	2026-10-17	450	Available	2
504	AB+	2026-09-06	2026-10-18	450	Available	2
505	O-	2026-09-08	2026-10-20	450	Available	1

Thank you for your attention.

Questions and discussion welcome